

K.R. Mangalam University

DSA Assignment:2

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B.TECH CSE(AI&ML)

Section: A

Submitted to:

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Task 1: Binary Search Tree (BST):

A Binary Search Tree (BST) is a tree data structure where each node has at most two children. The left subtree contains values smaller than the root, and the right subtree contains larger values, which helps in faster searching.

Operations:

- Insert elements
- Search elements
- Delete nodes (leaf, one child, two children)
- Inorder traversal

Explanation:

Inorder traversal follows Left → Root → Right, so it always prints elements in sorted order. This makes BST useful for maintaining sorted data.

Time Complexity:

- Average: $O(\log n)$
- Worst Case: $O(n)$ (when tree becomes skewed)

Task 2: Graph (BFS & DFS):

A graph is a collection of nodes connected by edges. It is used in real-world systems like social networks, maps, and routing. Graphs are commonly represented using adjacency lists.

BFS (Breadth First Search):

- Uses queue
- Traverses level by level

DFS (Depth First Search):

- Uses recursion/stack
- Goes deep before backtracking

Difference:

- BFS finds shortest path (unweighted graph)
- DFS is useful for cycle detection

Real-world Use:

- BFS → shortest path (navigation/maps)
- DFS → detecting loops in networks

Task 3: Hash Table:

A hash table stores key-value pairs and provides fast access using a hash function. It is widely used in searching and database indexing.

Collision:

When two keys map to the same index.

Solution:

Separate chaining (store multiple values in same bucket using list)

Time Complexity:

- Average: $O(1)$
- Worst: $O(n)$